

The Operating System Zoo

Overview

Operating systems are developed for different types of computers and different purposes. Some are built for large data centers, some for personal devices, and some for very small or special-purpose systems.

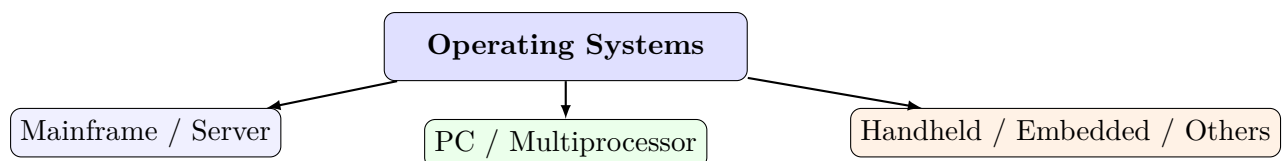
1. Main Types of Operating Systems

Categories

The main categories are:

- Mainframe OS
- Server OS
- Multiprocessor OS
- Personal Computer OS
- Handheld OS
- Embedded OS
- Sensor-Node OS
- Real-Time OS
- Smart Card OS

1.1 Simple Diagram



2. Mainframe Operating Systems

Definition

Mainframe operating systems run on very large computers in corporate data centers. They are designed to handle massive input/output and many jobs at once.

Key Points

- Used in high-end data centers
- Support batch processing
- Support transaction processing
- Support timesharing

Examples

OS/390, Linux, UNIX variants

3. Server Operating Systems

Definition

Server operating systems run on servers that provide services to many users over a network.

Functions

They provide services such as:

- file service
- print service
- web service

Examples

Solaris, FreeBSD, Linux, Windows Server

4. Multiprocessor Operating Systems

Definition

Multiprocessor operating systems are used in systems with multiple CPUs or multicore processors.

Key Points

- Support parallel processing
- Need communication and consistency mechanisms
- Important in modern multicore computers

Examples

Windows, Linux

5. Personal Computer Operating Systems

Definition

Personal computer operating systems are designed mainly for a single user.

Uses

They are used for:

- word processing
- spreadsheets
- games
- Internet access

Examples

Linux, FreeBSD, Windows, OS X

6. Handheld Operating Systems

Definition

Handheld operating systems run on smartphones, tablets, and similar portable devices.

Features

- portable design
- multicore support
- sensors, camera, GPS
- large number of apps

Examples

Android, iOS

7. Embedded Operating Systems

Definition

Embedded operating systems run inside devices like microwaves, TVs, cars, and music players.

Key Idea

These systems usually run fixed software stored in ROM, so users cannot normally install new applications.

Examples

Embedded Linux, QNX, VxWorks

8. Sensor-Node Operating Systems

Definition

Sensor-node operating systems run on tiny wireless sensor computers.

Characteristics

- battery-powered
- limited RAM
- event-driven
- designed for long unattended use

Example

TinyOS

9. Real-Time Operating Systems

Definition

Real-time operating systems are designed for systems where timing is critical.

Types

- **Hard real-time:** deadlines must always be met
- **Soft real-time:** occasional delay is acceptable

Examples

Hard real-time: industrial control, avionics
Soft real-time: multimedia, smartphones
OS example: eCos

10. Smart Card Operating Systems

Definition

Smart card operating systems run on tiny card-sized devices with very limited memory and power.

Key Points

- very small and simple
- some support only one function
- some can run Java applets
- need scheduling and protection for multiple applets

11. Conclusion

Summary

The operating system zoo includes many different types of systems. Each type is designed according to hardware size, user needs, and application purpose.